

Model Evaluation Report: main.py (v1)

1. Executive Summary

This report presents an assessment of the predictive accuracy of the **main.py (v1)** model, which has been trained on Welsh NHS pathway and activity data.

The model uses a **Random Forest regression approach** with time-series feature engineering (including lagged and rolling variables) to predict healthcare activity values across multiple pathways. Evaluation was conducted using a chronological train/test split to preserve time-order integrity.

Key performance metrics from the model evaluation are:

- Mean Absolute Error (MAE): **151,662.26**
- Root Mean Squared Error (RMSE): **217,659.53**
- Mean Absolute Percentage Error (MAPE): **291.81%**
- R^2 (coefficient of determination): **0.2263** [\[Results | Txt\]](#)

Overall, the model demonstrates **limited predictive accuracy**:

- The **R^2 value of 0.2263** indicates that the model explains only a small proportion of the variance in the target variable.
- The **very high MAPE (291.81%)** suggests substantial relative error, particularly where actual values are small or volatile.
- Absolute error measures (MAE and RMSE) are also large, reflecting scale and variability in the underlying data.

Feature importance analysis indicates that the model relies most heavily on:

- Short-term and medium-term historical values (rolling averages and lag features)
- Selected pathway-specific indicators (e.g. “patients waiting less than 26 weeks”) [\[Results | Txt\]](#)

The training dataset spans **monthly Welsh healthcare activity from January 2019 to December 2025**, covering diagnostics, emergency care, cancer pathways and waiting times. It includes significant structural changes (e.g. COVID-19 disruption), which introduce complexity and non-stationarity into the modelling problem.

[\[training_dataset.csv | Excel\]](#)

Conclusion:

The model captures some temporal patterns but does not provide sufficiently accurate or reliable predictions for operational or policy use without further development.

2. Background and Purpose

The purpose of this analysis is to **assess the predictive accuracy** of the model defined in **main.py (v1)** when applied to the provided training dataset.

The model is designed to:

- Learn patterns in historical Welsh NHS pathway data
- Predict future or unseen values in similar time-series contexts
- Support analytical understanding of healthcare demand and performance trends

The intended audience includes both **technical and executive stakeholders**, requiring clear interpretation of model performance and limitations.

3. Methodology Summary

Data Processing

- Input data loaded from training_dataset.csv
- Dates converted to datetime and sorted chronologically
- Non-numeric or missing values removed prior to modelling
[\[nhs_wales36...epoint.com\]](#)

Feature Engineering

The model constructs several time-series features:

- **Lag features:** 1, 2, 3, 6 and 12 months
- **Rolling averages:** 3-month and 6-month windows
- **Date features:** year and month
- **Categorical encoding:** one-hot encoding of description, geography, sex, age and source file [\[nhs_wales36...epoint.com\]](#)

Training Approach

- Dataset split chronologically (80% training / 20% testing)
- No random shuffling (to avoid future data leakage)

- Model: **RandomForestRegressor** with:
 - 300 trees
 - square-root feature selection
 - parallel processing enabled [\[nhswales36...epoint.com\]](#)

Evaluation Metrics

Performance was assessed using:

- Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - Mean Absolute Percentage Error (MAPE)
 - R^2 score [\[Results | Txt\]](#)
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4. Key Findings

4.1 Model Accuracy

- The model shows **low explanatory power ($R^2 = 0.2263$)**.
- **MAPE of 291.81%** indicates very high proportional error.
- Absolute errors (MAE and RMSE) are large, consistent with the wide range and variability of values across pathways. [\[Results | Txt\]](#)

Interpretation (Assumption):

High MAPE may be driven by:

- Low-value observations in some pathways
 - Structural shifts in time series (e.g. pandemic effects)
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4.2 Feature Importance

Top contributing features include:

- **rolling_6 (0.131813)**
- **rolling_3 (0.117988)**
- **lag_3 (0.110516)**
- **lag_12 (0.103388)**
- **lag_6 (0.101524)**

- Pathway-specific variables (e.g. “patients waiting less than 26 weeks”) [[Results | Txt](#)]

Interpretation:

- The model is primarily driven by **historical trends and momentum effects**
 - Structural categorical features contribute less than temporal signals
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4.3 Data Characteristics Influencing Results

The training data includes:

- Monthly healthcare activity across multiple pathways
- Significant **disruption during 2020 (COVID-19)**
- Long-term trends such as:
 - Increasing waiting times
 - Declining ED performance targets
 - Rising pathway delays [[training_dataset.csv | Excel](#)]

These features introduce:

- Non-linearity
 - Structural breaks
 - Heterogeneity across pathways
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5. Limitations and Caveats

This section reflects both **explicit limitations provided** and those directly observable in the modelling approach.

5.1 Data Limitations

- The dataset is **trimmed** (as stated in inputs), meaning:
 - Potential loss of early time-series context
 - Reduced ability to estimate long-term seasonality
 - Multiple pathways with very different scales are combined, increasing modelling complexity
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5.2 Model Limitations

- Random Forests do not inherently model:
 - Temporal continuity
 - Structural breaks or regime changes
 - No hyperparameter tuning beyond baseline settings is evidenced
 - No cross-validation beyond a single temporal split
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5.3 Evaluation Limitations

- High MAPE suggests instability in percentage-based error, particularly for:
 - Small denominators
 - Volatile series
 - No benchmarking against:
 - Simpler baselines (e.g. naive forecast)
 - Alternative models (e.g. ARIMA, gradient boosting)
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5.4 Structural Data Challenges

- The dataset includes:
 - Pandemic-driven shocks
 - Policy/system changes over time

Assumption:

These structural shifts reduce predictability and weaken model performance.

6. Recommendations

- [INFORMATION NEEDED: Confirmation whether recommendations are in scope of this request]
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7. Appendix

Model Evaluation Metrics

- MAE: **151,662.26**

- RMSE: **217,659.53**
- MAPE: **291.81%**
- R^2 : **0.2263** [\[Results | Txt\]](#)

Top 10 Feature Importances

1. rolling_6 – 0.131813
2. rolling_3 – 0.117988
3. lag_3 – 0.110516
4. lag_12 – 0.103388
5. lag_6 – 0.101524
6. patients waiting <26 weeks – 0.092985
7. lag_2 – 0.085183
8. lag_1 – 0.084742
9. source_file_rtt_patient_waiting_length.csv – 0.074543
10. year – 0.043922 [\[Results | Txt\]](#)

Dataset Overview

- Monthly Welsh healthcare data (2019–2025)
- Includes diagnostics, ED, cancer pathways and RTT waiting lists
- Contains pandemic-related disruptions and long-term trend changes